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Hallucination Detection Beyond Softmax

A Measurement-Space Impossibility and a Dual-NLI Protocol

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Abstract

Hallucination detectors for large language models read their evidence from natural-language inference (NLI) heads that are softmax-normalized over {entailment, neutral, contradiction}. Whether a detector uses both the entailment probability T and the contradiction probability F from one head, or uses a single channel whose complement is taken to be the other, the two signals live on a shared simplex and satisfy $T + F \leq 1$. We prove this as a one-line corollary of the softmax normalization (Theorem 1) and draw its consequence: the regime in which a text carries substantial evidence both for and against the same proposition, a graded analogue of the 'told-both' state of Belnap-Dunn logic, has no representation of its own: on the simplex it is confounded with weak evidence and with neutrality. The claim is representational, not decision-theoretic; whether a given benchmark penalizes the confound depends on its labels.

To recover the excluded regime we propose a dual-NLI protocol in which T and F come from two separately fine-tuned NLI models, and we probe it with public DeBERTa-v3-large and RoBERTa-large heads in four experiments on controlled, templated stimuli with oracle evidence spans (20 items per family, authored by us and released). (1) As an implementation check, single-NLI scoring escapes the simplex in 0/50 pairs, as it must. (2) A negative result: scoring a two-source premise holistically surfaces the conflict in only 4/20 items, because trained NLI models resolve internal conflict rather than report it, and in these stimuli they discount refuting content attributed to low-credibility sources. (3) Evidence decomposition, scoring T on the supporting span and F on the refuting one with a margin rule ($T + F > 1$ and $\min(T, F) \geq 0.15$), recovers 15/20 conflicting-evidence items. (4) Controls then show what the recovered signal is and is not: two sources that both support the claim are never flagged (0/20) and two that both refute it are flagged 1/20 times, but a refuting segment about a different claim is flagged 8/20 times, because the contradiction head fires on negation regardless of topic. Gating F with the entailment model's own neutrality removes those false positives but collapses recall to 5/20, since that model reads refutations from low-credibility sources as neutral: the gate re-imports the adjudication the protocol exists to avoid. A model-free lexical topicality gate keeps recall at 15/20; its false positives are 1/10 on entailment controls (one numeral paraphrase), 1/20 on agreeing refutations, and 0/20, 0/20 and 0/10 on the remaining families. Two further checks delimit the result. Swapping which model serves which channel drops recall to 4/20, so the recovery depends on the contradiction head used, not on the protocol alone; and since two calls of one head on two spans already leave the simplex, the escape is a property of decomposition, while the choice of head governs which behavioural failure is inherited: one head alone,

reading both spans, recovers 17/20. With oracle spans the support channel is nearly constant and the decision rule coincides with a gated threshold on the contradiction channel on three of four families; it does not on the fourth, where both sources refute the claim and the support channel is what blocks 17/20 false positives. We report the single-channel baseline explicitly.

The contribution is diagnostic: a measurement-space limitation stated precisely, a demonstration on controlled stimuli that decomposing evidence into separate NLI calls is what makes conflict representable, and controls that expose two behavioural counterparts of the constraint inside trained NLI heads, negation bias in one contradiction head and credibility discounting in another. Code, stimuli and raw scores are released.

Keywords: hallucination detection · large language models · natural language inference · softmax normalization · measurement-space impossibility · uncertainty quantification · conflicting evidence · calibration · paraconsistency.

1. Introduction

Hallucination, the production of confidently asserted but unsupported content, is the central deployment risk of large language models (LLMs) (Ji et al., 2023; Huang et al., 2023). The detection literature of the past three years has produced a family of techniques whose sophistication differs but whose measurement scaffolding is uniform: each output, or each atomic claim within it, is mapped to a scalar in [0, 1], typically through softmax-normalized NLI probabilities, and a binary {factual, hallucinated} label is recovered by thresholding.

The surveys of that literature (Ji et al., 2023; Huang et al., 2023; Tonmoy et al., 2024) agree that no detector approaches ceiling performance and offer no structural account of why. We propose a partial one. The measurement space implicit in current methods has no representation for a state that retrieval-augmented pipelines routinely encounter: the state in which the available evidence supports and contradicts the same proposition at once. When both the truth-supporting score T and the truth-contradicting score F are read from one softmax-normalized NLI head, or when only one of them is read and the other is its implicit complement, the constraint $T + \text{neutral} + F = 1$ implies $T + F \leq 1$. The regime $T + F > 1$ is unreachable.

This note makes the limitation explicit, proves it, shows with real NLI models that the excluded region is populated once evidence is decomposed, and supplies an evidence-decomposed dual-NLI protocol together with the controls that show it measures conflict rather than an artifact of its own construction.

1.1 Contribution

1. Theorem 1 (Softmax-Paraconsistency Impossibility). If a detector's only evidence features are the entailment and contradiction probabilities of one softmax-normalized NLI call, then $T + F \leq 1$ and the state of simultaneous support and contradiction has no representation distinct from partial evidence or neutrality. The result is elementary; its content is the explicit statement of an assumption the detection literature has not examined.
2. Diagnostic mapping. We state precisely which quantity three families of detectors (semantic entropy, SelfCheckGPT, NLI-based fact verification) actually read, and show that each lives on a shared simplex.
3. Dual-NLI protocol. T and F are computed by two separately fine-tuned NLI models, with evidence decomposition and a margin decision rule.

4. A diagnostic with real NLI models, oracle spans and controls. Naive holistic dual-NLI fails informatively (4/20); decomposition recovers 15/20 conflicting-evidence items; three decomposition controls and two gating variants show that the ungated contradiction channel is negation-biased and that an NLI-based gate inherits credibility discounting, while a lexical gate keeps 15/20 recall at 2/80 false positives. The margin threshold is characterized by a sweep rather than fixed post hoc.
5. Open-source reference implementation, stimuli and raw scores released under MIT License.

1.2 Scope

This is a measurement note. It establishes a limitation, a protocol and its controls on curated stimuli scored by public NLI models. It does not estimate how often frontier LLM outputs occupy the excluded regime on standard benchmarks; that is a corpus question, addressed in follow-up work (Section 7). Releasing the theorem and the protocol separately lets the formal claim be scrutinized without being conditional on any one benchmark design.

2. Background

2.1 What current detectors actually measure

We examine three families of detectors and state, for each, which quantity is read from the NLI head. The point is not that every method extracts both T and F; it is that none of them measures a contradiction channel that is free to vary independently of the support channel.

Semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024) clusters stochastic samples of an LLM answer by bidirectional entailment, decided by a single softmax-normalized NLI model (DeBERTa-v3-large-MNLI), and treats the entropy over clusters as the hallucination signal. The entropy itself is computed over many samples and is not a point on any NLI simplex; the constraint enters one level down, in the pairwise equivalence decision, which is an argmax on the shared simplex. Two samples that support and contradict each other's content at once are assigned to one cluster or to two; the method has no third option, and the aggregate cannot recover what the pairwise decision discarded.

SelfCheckGPT (Manakul et al., 2023) has several variants (BERTScore, question answering, n-gram, prompting); only the NLI variant falls under our analysis. It scores a sentence by the contradiction probability of a single RoBERTa-large-MNLI head against each resample. Only F is read; the complement $1 - F$ is non-contradiction, which pools entailment with neutral. The measurement is one-dimensional, and support and contradiction cannot both be high.

NLI-based fact verification (Laban et al., 2022; Min et al., 2023) decomposes a response into atomic claims and verifies each against retrieved evidence. SummaC reads entailment scores from one NLI head; FActScore, in its standard configuration, elicits a binary supported/not-supported verdict from a language model, which is a scalar with implicit complement. Calibration studies that build on these signals (Tian et al., 2023; Xiong et al., 2024; Kadavath et al., 2022) inherit the same one-dimensional or shared-simplex measurement.

In every case T and F are either read from one softmax call or reduced to one scalar and its complement. This is inheritance rather than design: MNLI and FEVER are trained as three-class classification problems with cross-entropy loss, which yields softmax outputs by construction, and downstream pipelines consume those outputs as given. Two qualifications bound the scope. Semantic entropy is not a conflicting-evidence detector and does not aim to be one; what we claim is only that the pairwise NLI call it relies on carries the constraint. And detectors that do not use NLI probabilities at all, such as question-generation-and-answering metrics (Honovich et al., 2021; Fabbri et al., 2022), which compare answers by string overlap, fall outside Theorem 1; whether they can represent conflicting evidence is a separate question that we do not address.

2.2 Paraconsistency and formal antecedents

The regime $T + F > 1$ is what formal logic calls paraconsistency: both a proposition and its negation are supported without explosive trivialization (da Costa, 1974; Priest, 1979; Carnielli & Coniglio, 2016). Belnap-Dunn four-valued logic (Belnap, 1977) names the four epistemic states told-true, told-false, told-both and told-neither. We use the graded analogue in which T and F are independent degrees in [0, 1]: the four Belnap states are the corners (1,0), (0,1), (1,1) and (0,0) of the unit square, and $T + F > 1$ is the half of the square nearer to the told-both corner than to told-neither. The inequality is thus the reachability criterion; the operational reading of told-both used in our decision rule additionally requires both channels to be active, $\min(T, F) \geq \tau$, since a point such as (0.97, 0.05) is above the diagonal without carrying any opposing evidence. Neutrosophic logic (Smarandache, 1998; Wang et al., 2010) extends the picture to a continuous triplet (T, I, F) in $[0, 1]^3$ with no sum constraint, and our earlier work documented the corresponding behaviour in LLM self-reports (Leyva-Vázquez & Smarandache, 2026). The present paper supplies the operational bridge from these frameworks to the NLI measurement pipeline.

2.3 Relation to uncertainty quantification

Within machine-learning uncertainty quantification, the closest structural analogues of the dual-NLI regime are credal sets (Hüllermeier & Waegeman, 2021) and deep ensembles (Lakshminarayanan et al., 2017). A credal set represents epistemic uncertainty as a convex set of distributions; the told-both state corresponds to a credal set containing distributions that place high mass on entailment and on contradiction respectively. Deep ensembles aggregate independently trained models to capture diversity a single head cannot. The dual-NLI protocol is related to both but targets a different object: not predictive variance, but a region of the measurement space that a single head cannot reach. Conformal methods (Angelopoulos & Bates, 2023) provide coverage guarantees on top of a given score and do not relax the simplex constraint of the score itself; our correction is complementary to them.

Evidential deep learning (Sensoy et al., 2018) deserves a separate remark because it is often presented as the answer to softmax overconfidence. An evidential classifier outputs Dirichlet concentration parameters whose subjective-logic reading assigns a belief mass b_k to each class and an uncertainty mass u, with Σ b_k + u = 1. The entailment and contradiction beliefs therefore satisfy b_T + b_F $\leq 1 - u \leq 1$: evidential outputs separate lack of evidence from evidence, which softmax cannot, but they still cannot represent evidence for both classes that sums above one. In Dempster-Shafer terms, they place no mass on the joint set {entailment, contradiction}. The shared-simplex constraint of Theorem 1 is thus not specific to softmax; it is a property of any head whose class masses are normalized together, and evidential heads inherit it.

2.4 Closest neighbours and what this paper adds

Three lines of work come closest and should be distinguished explicitly. Mündler et al. (2023) detect self-contradiction between two statements produced by the same model; the object measured is inconsistency between outputs, not the coexistence of support and refutation for one claim in the evidence. Shi et al. (2026) decompose the uncertainty of vision-language models into belief divergence and belief conflict using Dempster-Shafer theory over multiple sampled answers; conflict is again measured across outputs, whereas we measure it across evidence sources for a fixed claim, and we locate the obstacle in the NLI measurement head rather than in the sampling scheme. Galitsky (2026) detects unsupported or contradictory content in explanations through an information-theoretic abduction model; his contradiction signal is still read from a scalar consistency score. None of the three states the simplex constraint or proposes to measure T and F on separate heads.

The human-label-variation literature is adjacent but must not be over-read. ChaosNLI (Nie et al., 2020) collects 100 annotations per item on SNLI and MNLI and finds items on which annotators split between entailment and contradiction; Baan et al. (2022) show that calibrating to a majority label is ill-posed on such items, and Plank (2022) reviews the phenomenon across tasks. Each annotator, however, emits one exclusive label, so the vote shares of an item still sum to one: ChaosNLI documents a mixture of readings across people, not a single judge who holds both. What it does establish is that ordinary NLI items exist for which both the entailment and the contradiction reading are live, which is the situation a two-source premise makes explicit. Whether individual humans, asked for separate support and refutation ratings, occupy $T + F > 1$ is an open empirical question that our stimuli do not address.

3. The Softmax Trap

3.1 Theorem and proof

Theorem 1 (Softmax-Paraconsistency Impossibility). Let M be a softmax-normalized NLI model emitting probabilities for three classes {entailment, neutral, contradiction} for a premise-hypothesis pair (p, h). Define T(p, h) = entailment(p, h) and F(p, h) = contradiction(p, h). Then for all pairs (p, h):

$$\mathrm{T(p, h) + F(p, h)} \leq 1, \quad (1)$$

with equality if and only if neutral(p, h) = 0.

Proof. By the definition of softmax,

$$\text{entailment} + \text{neutral} + \text{contradiction} = 1, \quad \text{each} \in [0, 1]. \quad (2)$$

Substituting T = entailment, F = contradiction:

$$\mathrm{T + F = 1 \text{ - } neutral} \leq 1. \quad (3)$$

Equality holds when neutral = 0. ■

Corollary 1 (Representational limitation). Assume a detector whose only evidence features for a (premise, claim) pair are the entailment and contradiction probabilities of one softmax-normalized NLI call, and define the simultaneous-evidence state as $T + F > 1$. Then the detector assigns no input to that state. In particular, a premise that supports and refutes the claim is mapped to the same region of the simplex as a premise carrying weak or partial evidence, or none: the representation is many-to-one on exactly the cases of interest. The corollary says nothing about detectors that use further features (retrieval structure, multiple NLI calls, logits, source metadata); for those, the question is whether any such feature restores the distinction, which is what the dual-NLI protocol does explicitly.

Remark (single-channel detectors). Detectors that read only one channel, typically F, and treat non-contradiction $1 - F$ as their positive signal are subject to the same limitation, but for a different and trivial reason: the two numbers sum to one by definition. Theorem 1 is not needed for them. We mention them because they are the majority of deployed NLI-based detectors and because the point of Section 4 is the same for both groups: a second, independently measured channel is required before the told-both state can be represented at all.

Remark (what the constraint is about). The point (T, neutral, F) = (0.5, 0, 0.5) is reachable, and one may ask whether it is not already a representation of told-both. It is not, and the reason is the content of the theorem: a softmax head reports a posterior over three mutually exclusive relations, so (0.5, 0, 0.5) says that the relation is entailment or contradiction with equal probability and is certainly not neutral. That is the coordinate a head emits for an item on which it cannot decide between two exclusive readings. The state in which both relations hold to a high degree is a different epistemic situation and receives the same coordinates. What $T + F > 1$ encodes, once T and F are measured by separate calls, is precisely the failure of exclusivity: two degrees that are free to be high together. The theorem is elementary; its interest lies in the fact that the exclusivity assumption is never stated in the detection literature and is inherited by every pipeline that consumes the probabilities of one call.

Remark (per call, not per model). Theorem 1 binds one softmax call on one (premise, claim) pair. Two calls of the same head on two different spans of the premise are not bound by it: entailment_A(span_1, h) and contradiction_A(span_2, h) may sum above one. The escape from the simplex is therefore a property of decomposing the evidence into separate calls, not of using two models. Using two models is a further design choice, and Experiments 5 and 6 test what it changes.

3.2 Geometric interpretation

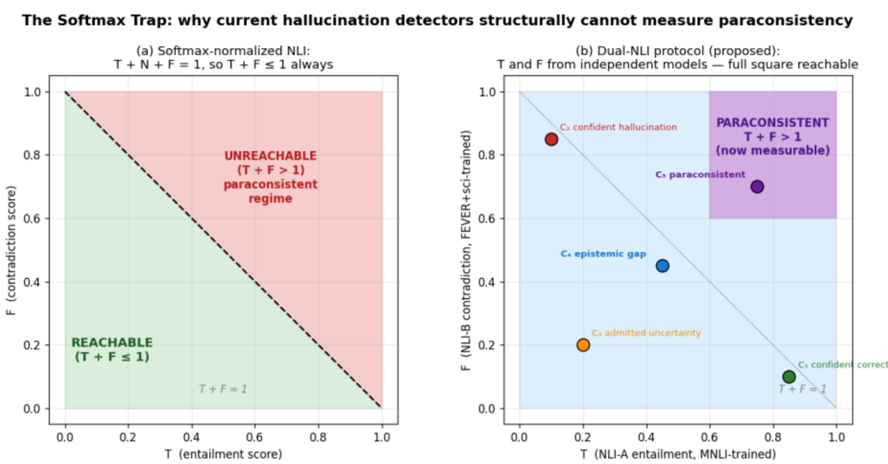

Figure 1. (a) Under softmax-normalized NLI the reachable region of (T, F) is the lower triangle $T + F \leq 1$; the upper triangle is unreachable by construction. (b) With T and F from independent models the full unit square becomes reachable.

The reachable region under a shared softmax is the lower triangle of the unit square, a set of measure 1/2. The excluded region has equal measure and is structurally inaccessible: no additional training data, model scale or calibration moves the boundary.

3.3 Why this bears on the detection ceiling

If some fraction of the items a detector faces carry conflicting evidence, a detector confined to the simplex must place them somewhere on the simplex: it resolves the conflict in one direction or calls it neutral. Whether that counts as an error depends on the reference labels. Under a binary {supported, unsupported} scheme it may not; under any scheme that records conflict, or in any downstream use that should treat contested claims differently from unknown ones, it does, and the error is a consequence of the measurement space rather than of labelling noise. The size of that fraction on standard benchmarks is unknown. We establish here only that the geometry lacks the state and that a protocol exists which admits it.

4. Dual-NLI Protocol

4.1 Definition

To remove the constraint while preserving the rest of the pipeline, T and F are computed by two separate softmax heads belonging to two differently fine-tuned NLI models:

$$T(p, h) = \text{entailment\_A}(p, h), \quad F(p, h) = \text{contradiction\_B}(p, h), \quad (4)$$

where A is a general-domain entailment model and B a model trained with additional contradiction-rich data. Each remains softmax-normalized internally, but T and F no longer share a simplex, and $T + F > 1$ becomes representable. The architecture resembles a two-member deep ensemble (Lakshminarayanan et al., 2017) except that the members are not averaged: each contributes a structurally distinct channel. An indeterminacy channel I, when wanted, is computed separately from self-consistency across resamples and hedging markers; it plays no role in the results below.

4.2 Independence assumption

Theorem 1 is escaped as soon as no shared softmax couples T and F; statistical independence of A and B is not required and does not hold, since both are trained on overlapping NLI corpora. The empirical question is whether the joint distribution of (T_A, F_B) is dispersed enough to populate the excluded regime on realistic inputs, and whether it does so only when the evidence is in fact conflicting. Sections 5.2 and 5.3 test both halves.

4.3 Computational cost

The protocol doubles NLI inference (two forward passes per pair). With evidence decomposition the passes are shorter, since each model reads one segment rather than the concatenation.

4.4 Evidence decomposition and the margin rule

Two refinements complete the protocol. Evidence decomposition: when the premise aggregates material from more than one source, T is scored on the segment(s) that support the hypothesis and F on the segment(s) that oppose it, rather than on the concatenated premise. In this paper the segments are oracle spans, known by construction; the protocol as evaluated here is therefore a measurement diagnostic, and an end-to-end detector would need a span-finding stage that we do not build. Trained NLI models are optimized to output one label per (premise, hypothesis) pair; given an internally conflicting premise they resolve the conflict instead of reporting it, so the conflict must be exposed at the input. Margin rule: an item is flagged as simultaneous-evidence iff $T + F > 1$ and $\min(T, F) \geq \tau$. The margin removes boundary artifacts in which a near-certain single channel plus channel noise sums marginally above 1. We report $\tau = 0.15$ in the headline numbers and characterize the choice with a sweep in Section 5.3.

A third element is forced by the controls of Section 5.3: a topicality gate on the F channel. Contradiction heads fire on negated sentences whether or not the sentence is about the claim, so F must be kept only when the refuting segment is on-topic. We test two gates. An NLI gate uses Model A's neutral probability on (refuting segment, claim) as an off-topic score, $F_{nli} = F_B \cdot (1 - \text{neutral}_A)$. A lexical gate uses content-word overlap, $F_{lex} = F_B$ if $J(\text{segment, claim}) \geq \theta$ else 0, with J the Jaccard index over stemmed content words after removing stopwords, negation markers and source-attribution vocabulary. The gates differ in what they import, and Section 5.3 shows that the difference matters. We are explicit about the status of the lexical gate: it is a diagnostic that isolates the topicality requirement on stimuli where refutations restate the claim's content words, not a component we propose for in-the-wild text, where refutations are paraphrased.

5. Empirical Validation with Real NLI Models

5.1 Setup

Model A (T channel) is MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli (He et al., 2021, fine-tuned on MNLI, FEVER-NLI, ANLI, LingNLI and WANLI); Model B (F channel) is ynie/roberta-large-snli_mnli_fever_anli_R1_R2_R3-nli. The two share no softmax. All runs execute on CPU in minutes and are scripted in the repository.

- Experiment 1 (theorem confirmation and stimulus lesson). The 50 pairs of the earlier heuristic pre-study: 10 entailment, 10 contradiction, 10 neutral and 20 items whose hypothesis qualifies itself ("X, but it is also not X"), scored under dual-NLI and under single-NLI (both channels from Model A's softmax).
- Experiment 2 (conflicting evidence, holistic vs. decomposed). Twenty items in which the conflict lives in the evidence: each premise concatenates a supporting source and a refuting source, and the refuting source restates the claim under negation (item 2, verbatim: "The physics textbook states that water boils at 100 degrees Celsius at sea level. However, the lab report concludes that water does not boil at 100 degrees Celsius at sea level."); the hypothesis is the atomic claim. All 20 items are listed in Appendix A. Each item is scored holistically (full premise to both models) and decomposed (supporting segment to A, refuting segment to B).
- Experiment 3 (controls for decomposition). For the same 20 claims, three control families scored with the identical decomposed protocol: AGREE-SUP, two independent sources that both support the claim (source 1 to A, source 2 to B); AGREE-REF, two sources that both refute it; CROSS, the supporting segment of claim i paired with the refuting segment of a different claim j. If decomposition flagged conflict merely because two clear segments were handed to two models, these controls would be flagged; they should not be.
- Experiment 4 (topicality gates). The conflicting items, the three control families and the single-segment controls rescored with the NLI gate and with the lexical gate of Section 4.4 ($\theta = 0.3$), reporting recall and false positives under the same margin rule.
- Experiment 5 (model-assignment swap). The decomposed scoring of Experiments 2-3 repeated with the channel roles exchanged (T from Model B, F from Model A), to check whether recovery is a property of one pairing.
- Experiment 6 (single-head baselines and F-only rule). Decomposed scoring with both channels from Model A alone and from Model B alone, which Theorem 1 does not forbid, and the single-channel rule $F \geq 0.15$ on the gated refuting span for every configuration, since with oracle spans the support channel is nearly constant.
- Single-segment controls. The 10 entailment and 10 contradiction pairs of Experiment 1, scored under dual-NLI (the premise serves as both segments).

All stimuli are English sentences over encyclopedic facts, authored by us and released verbatim; the refuting sources deliberately vary in attributed credibility (textbook, lab report, pamphlet, blog), which turned out to matter.

5.2 Results: impossibility, negative result, recovery

Implementation check. Under single-NLI scoring no pair among the 50 of Experiment 1 produces $T + F > 1$ (0/50). This is not evidence for Theorem 1, which is an identity; it is a unit test that the scoring code reads the head it claims to read, and we keep it in Table 1 for that reason only.

A stimulus lesson. The 20 self-qualifying items of Experiment 1 are not recovered by dual-NLI either (0/20 raw; mean $T = 0.038$, mean $F = 0.010$): both models read a hypothesis that qualifies itself as neither entailed nor contradicted by a simple premise. This exposes the earlier heuristic recovery of the v0.5 preprint as an artifact of token-overlap scoring, and it relocates the phenomenon: conflict must live in the evidence, not in the hypothesis.

A negative result. Scored holistically, the corrected items of Experiment 2 cross the diagonal in only 4/20 cases (mean $T + F = 0.713$; 3/20 under the margin rule). Inspection shows why: the models resolve the conflict rather than report it, and they use source-credibility cues to do so. Refuting content attributed to a "disputed pamphlet" or a "fringe theory" is discounted almost entirely. This is what a cross-entropy-trained three-way classifier is built to do: emit one label per pair. Given a premise that contains both, it picks a side, and the conflict is lost before any downstream method sees it.

Decomposition recovers the regime. Under evidence-decomposed scoring the same 20 items reach mean $T + F = 1.350$, with 16/20 above the diagonal and 15/20 (75%) under the margin rule. On the single-segment controls the raw rule $T + F > 1$ misfires on 8/10 entailment pairs ($T = 0.997$ plus channel noise $F = 0.053$), a boundary artifact that the margin removes (1/20 false positives).

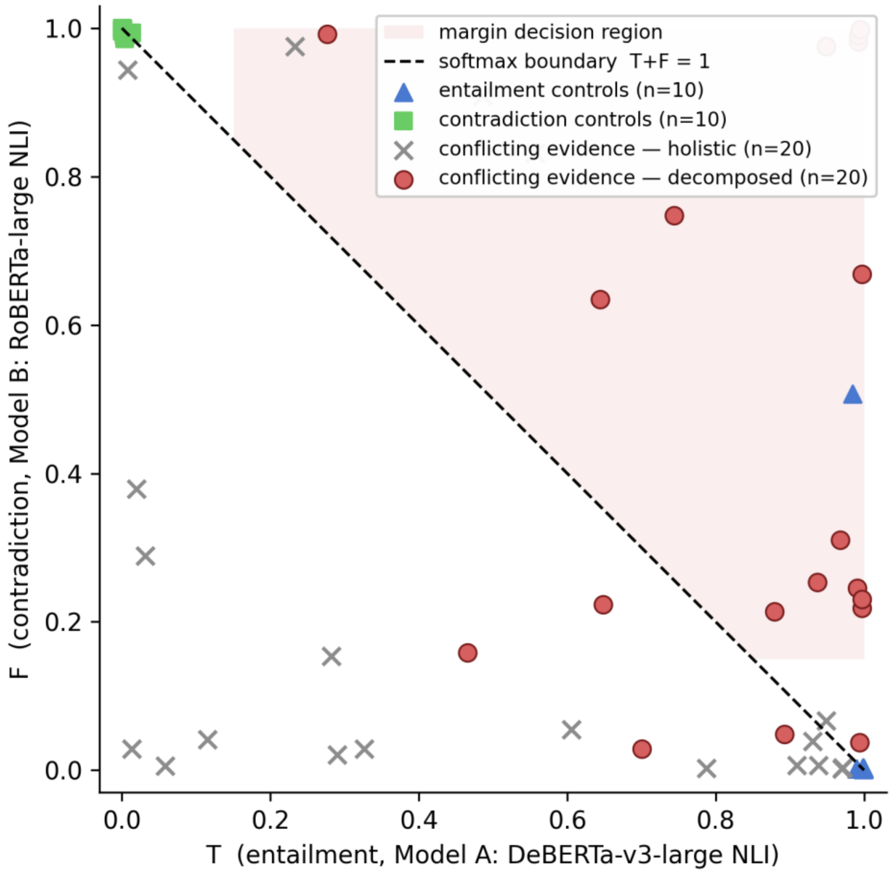

Figure 2. Real-model scores in the (T, F) unit square. The dashed line is the softmax boundary T + F = 1. Holistic scoring of conflicting-evidence items (crosses) stays largely inside the simplex; evidence-decomposed scoring (filled circles) populates the excluded region. Entailment and contradiction controls sit on their edges; the shaded band is the margin decision region.

Setting	n	mean T	mean F	mean T+F	raw T+F>1	margin rule
E1 all pairs, single-NLI	50	-	-	-	0/50	0/50
E1 self-qualifying, dual-NLI	20	0.038	0.010	0.048	0/20	0/20
E2 conflicting, holistic	20	0.474	0.239	0.713	4/20	3/20
E2 conflicting, decomposed	20	0.853	0.498	1.350	16/20	15/20
Entailment controls, dual-NLI	10	0.997	0.053	1.049	8/10	1/10
Contradiction controls, dual-NLI	10	0.003	0.995	0.998	2/10	0/10

Table 1. Experiments 1 and 2 (Model A: DeBERTa-v3-large NLI; Model B: RoBERTa-large NLI). The margin rule flags an item iff T + F > 1 and min(T, F) ≥ 0.15. Per-item scores are released as CSV.

5.3 Results: what the decomposed signal measures

The recovery in Section 5.2 would be uninformative if any two clear segments handed to two models produced T + F > 1. Experiment 3 tests this with the same claims and the same protocol (Table 2). Two of the three controls behave as they should. When both sources support the claim (AGREE-SUP), Model B reads the second supporting source and returns essentially no contradiction (mean F = 0.002; 0/20 flagged). When both refute it (AGREE-REF), Model A returns low entailment (mean T = 0.179; 1/20 flagged). Agreement is not mistaken for conflict.

The third control fails, and the failure is informative. When the supporting segment of claim i is paired with the refuting segment of a different claim j (CROSS), Model B still returns substantial contradiction (mean F = 0.284) and 8/20 items are flagged. The contradiction head fires on a negated sentence regardless of what it negates. The ungated F channel therefore measures, in part, the presence of a refutation rather than a refutation of this claim. This is the second behavioural counterpart of Theorem 1 we document: the first (Section 5.2) was that a shared head adjudicates conflict; this one is that a separated contradiction head over-generalizes negation.

Experiment 4 asks whether a topicality gate can remove the CROSS false positives without destroying recall (Table 3, Figure 3). The NLI gate does the first and not the second: CROSS drops to 0/20, but recall on genuine conflicts collapses from 15/20 to 5/20. The reason is visible in the gate itself: Model A's mean neutral probability on the refuting segment is 0.912 for CROSS, as intended, but also 0.635 for genuine refutations, because Model A reads a refutation attributed to a pamphlet or a blog as neither entailing nor contradicting the claim. The gate re-imports the source-credibility adjudication of Section 5.2 through the back door. The lexical gate does not consult any NLI model; it separates the families cleanly (mean J = 0.85 for conflicting items against 0.01 for CROSS) and leaves recall at 15/20 with CROSS at 0/20. The remaining false positives under the lexical gate are 1/10 on entailment controls, a single pair in which Model B reads a numeral paraphrase ('one hundred' vs '100') as contradiction, and 1/20 on AGREE-REF.

Family (margin rule, τ = 0.15)	**n**	**ungated F_B**	**NLI gate**	**lexical gate (θ = 0.3)**
Conflicting evidence (recall)	20	15/20	5/20	15/20
AGREE-SUP (false positives)	20	0/20	0/20	0/20
AGREE-REF (false positives)	20	1/20	0/20	1/20
CROSS (false positives)	20	8/20	0/20	0/20
Entailment controls (false positives)	10	1/10	1/10	1/10
Contradiction controls (false positives)	10	0/10	0/10	0/10

Table 3. Experiment 4: recall and false positives of the decomposed protocol without a gate, with the NLI gate F_B · (1 − neutral_A) and with the lexical gate. Same margin rule throughout.

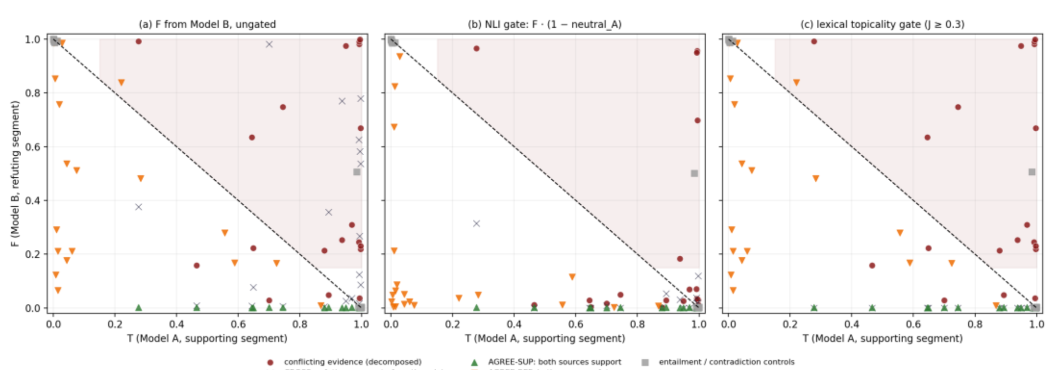

Figure 3. Conflicting items and the five control families in the (T, F) square (entailment and contradiction controls share one marker): (a) ungated, (b) NLI gate, (c) lexical gate. The NLI gate empties the CROSS crosses from the flagged region but drags most genuine conflicts (circles) below the diagonal; the lexical gate removes the crosses and leaves the circles where they were.

Control family (decomposed scoring)	n	mean T	mean F	mean T+F	raw T+F>1	margin rule
AGREE-SUP: two sources support	20	0.853	0.002	0.854	0/20	0/20
AGREE-REF: two sources refute	20	0.179	0.481	0.661	3/20	1/20
CROSS: support i + refute j≠i	20	0.853	0.284	1.136	12/20	8/20
E2 conflicting, decomposed (reference)	20	0.853	0.498	1.350	16/20	15/20

Table 2. Experiment 3: the decomposed protocol applied to non-conflicting two-source premises and to cross-item pairings. The conflicting-evidence row of Table 1 is repeated for reference.

Threshold sensitivity and what was tuned on what. The margin $\tau = 0.15$ was chosen on the single-segment entailment and contradiction controls; those 20 items are therefore a tuning set, and their 1/20 in Tables 1 and 3 is reported for completeness, not as evidence. The conflicting items and the three decomposition families were not consulted when τ was fixed, and Table 4 reports the full sweep on all of them. Recall on conflicting items is flat up to $\tau = 0.20$ and degrades beyond; AGREE-SUP is zero at every τ; the CROSS column is the one τ cannot fix, which is why a topicality gate is needed rather than a larger margin. The lexical threshold θ was not tuned at all: the Jaccard index of every conflicting item is at least 0.50 and that of every CROSS item at most 0.125, so any θ in (0.125, 0.50) gives the numbers of Table 3 (we report $\theta = 0.3$; 0.2, 0.4 and 0.5 give identical counts).

τ	recall, E2 decomposed	FP, ENT/CON	FP, AGREE-SUP	FP, AGREE-REF	FP, CROSS
0.00	16/20	10/20	0/20	3/20	12/20
0.05	15/20	1/20	0/20	1/20	10/20
0.10	15/20	1/20	0/20	1/20	9/20
0.15	15/20	1/20	0/20	1/20	8/20
0.20	15/20	1/20	0/20	1/20	8/20
0.25	11/20	1/20	0/20	0/20	8/20
0.30	9/20	1/20	0/20	0/20	7/20
0.40	8/20	1/20	0/20	0/20	6/20

Table 4. Sweep of the margin threshold τ in the rule $T + F > 1$ and $\min(T, F) \geq \tau$, ungated F.

Model-assignment swap. To check that the recovery is not a property of the particular pairing, Experiment 5 repeats the decomposed scoring with the roles exchanged (T from Model B's entailment, F from Model A's contradiction). With the lexical gate and the same margin rule the swapped protocol recovers 4/20 conflicting items and flags 0/20 AGREE-SUP, 0/20 AGREE-REF and 0/20 CROSS items (ungated CROSS: 0/20). The swap removes the CROSS false positives without any gate and loses most of the recall: Model A's contradiction head discounts refutations attributed to weak sources, the same behaviour that defeated holistic scoring in Section 5.2, while Model B's fires on negation regardless of topic. The recovery therefore belongs to the pairing, not to the protocol in the abstract, and the honest summary is that the two public heads we tested fail in opposite directions. Neither assignment was tuned; we report both.

Single-head baselines and the one-channel rule. Because Theorem 1 binds a call and not a model, the natural baseline is decomposition with one head serving both channels (Table 5). With oracle spans the support channel barely varies (mean $T_A = 0.853$ on conflicting items and 0.853 on CROSS, because the supporting span is the same). Table 5 therefore reports the single-channel rule $F \geq 0.15$ on the gated refuting span beside the two-channel rule. On conflicting items, AGREE-SUP and CROSS the two rules coincide, as the constancy of T predicts. They come apart exactly on AGREE-REF, where both spans refute the claim: the single-channel rule flags 17/20 of those items as conflict, the two-channel rule 1/20, because T is low when no span supports the claim. The support channel is what distinguishes contested from unanimously refuted, and that is its job. Second, the escape from the simplex needs no second model: Model B alone, reading both spans in two calls, recovers 17/20 conflicting items with the gate and 0/20 CROSS false positives, marginally more recall than the dual assignment and 4/20 rather than 1/20 AGREE-REF false positives. Model A alone recovers 4/20. What governs the outcome is which head reads the refuting span, not how many heads there are.

Configuration (gated, $\tau = 0.15$)	Conflicting: margin | F-only	AGREE-SUP	AGREE-REF	CROSS
A only (T_A, F_A)	4/20 | 4/20	0/20 | 0/20	0/20 | 3/20	0/20 | 0/20
B only (T_B, F_B)	17/20 | 17/20	0/20 | 0/20	4/20 | 17/20	0/20 | 0/20
dual, A→T, B→F (main)	15/20 | 17/20	0/20 | 0/20	1/20 | 17/20	0/20 | 0/20
dual, B→T, A→F (swap)	4/20 | 4/20	0/20 | 0/20	0/20 | 3/20	0/20 | 0/20

Table 5. Experiment 6. Each cell gives counts flagged by the two-channel margin rule | by the single-channel rule $F \geq 0.15$, both with the lexical gate. Configurations: one head for both channels (A only, B only), the main dual assignment and its swap.

Independence diagnostic. Pearson correlation between T_A and $1 - F_B$ is $r = 0.285$ over the 50 pairs of Experiment 1 and $r = -0.392$ over the 30 neutral and self-qualifying pairs, where the correlation is not imposed by the class. Neither approaches the $r = 1$ of softmax coupling; the negative value on the class-free subset says only that on ambiguous inputs the two heads do not move together.

5.4 Interpretation

Three lessons. First, the impossibility half of the paper is empirical as well as formal: real single-NLI scoring produced no escape from the simplex in 50 attempts. Second, reachability of the excluded regime is a property of the measurement protocol, not of the two-model architecture alone: naive dual-NLI fails because the component models were trained to adjudicate conflict, including by source-credibility heuristics, so the conflict must be exposed at the input. We regard this negative result as informative: the algebraic constraint of Theorem 1 has a behavioural counterpart inside trained NLI models. Third, the controls of Experiments 3 and 4 delimit what the recovered signal is: two agreeing sources never produce it; a refutation of another claim does, unless the F channel is gated for topic; and gating with the entailment model's own judgement brings the adjudication problem back. The clean version of the protocol is therefore decomposition plus a model-free topicality gate plus a margin; its recall on curated conflicts is 15/20, not 100%, and it stays silent on 78/80 control items across the five families.

6. Discussion

6.1 Why the field missed this

The softmax constraint is embedded in standard NLI training and inference and operates as an unstated assumption. None of the surveyed detection methods discusses it. We hypothesize that it was inherited from the three-class framing of MNLI and propagated through the literature without re-examination, a familiar pattern in which constraints embedded in tooling propagate downstream until someone formalizes them.

6.2 Implications for evaluation

1. Reported detection scores are measured inside a space that cannot represent conflicting evidence. Comparisons across methods remain valid; the interpretation of the residual error as noise does not, until the conflicting fraction of the benchmark has been measured.
2. Calibration methods that train on softmax-normalized NLI features inherit the constraint and cannot remove it by better calibration; a protocol-level change is required.
3. Any pipeline that aggregates atomic-claim verdicts (retrieval-augmented fact checking, rubric-based judging) compounds the constraint at each claim, so the blind spot grows with the number of claims aggregated.

6.3 Credal reading

In the credal-set formalism (Hüllermeier & Waegeman, 2021) the dual-NLI protocol extracts two boundary representatives, the most-entailing and the most-contradicting model, whose joint output spans the unit square rather than the simplex. This is a practical approximation to credal scoring for NLI at twice the inference cost of a single head, and the margin rule is a crude but explicit decision boundary on that square.

7. Limitations and follow-up

- Theorem 1 covers detectors that read T and F from one head or reduce them to one scalar; detectors that already combine several heads or hand-crafted features fall outside it and should be examined case by case.
- Evidence decomposition presupposes a segmentation of the premise into supporting and refuting spans. In our stimuli the spans are oracle spans, given by construction. This makes Experiments 2-4 a diagnostic of the measurement head under known conflict, not an evaluation of an end-to-end detector; finding the spans is part of the detection problem and is untested here.
- The source-credibility reading of the holistic failure is a hypothesis, not an isolated effect: the attributions (pamphlet, blog, lab report) are confounded with lexical content and item. A factorial design crossing topic relevance, explicit negation, lexical overlap and attribution is needed before either the credibility effect or the negation bias is asserted as a general property of NLI heads; what we report is what these two heads did on these items.
- Twenty conflicting-evidence items and eighty control items in English over encyclopedic facts, all following one template ("Source A states X. However, Source B concludes not-X."), constitute a controlled probe, not a corpus study. The counts in Tables 1-4 are exact for these stimuli and should not be read as rates for any population of texts; a corpus study with naturally occurring conflicting

evidence (for example, contested items of ChaosNLI or retrieval-augmented QA with disagreeing passages) is required before any rate is claimed.

- The source-credibility effect is documented but not isolated; disentangling it from lexical confounds requires a dedicated design.
- The margin τ is characterized by a sweep, not derived; a principled calibration remains open.
- The lexical topicality gate is a surface heuristic that works on stimuli where the refuting segment restates the claim's content words; on paraphrased or cross-lingual refutations it will under-fire. A semantic-similarity gate that does not pass through an NLI head is the natural replacement and is untested here.
- One entailment control (a numeral paraphrase) is flagged under every setting; with 10 such controls this is a 10% false-positive rate on paraphrase pairs and should be read as such.
- The rate of simultaneous-evidence behaviour in frontier-LLM outputs on TruthfulQA (Lin et al., 2022), HaluEval (Li et al., 2023) and FActScore is unknown and is the object of follow-up work; nothing in the present results depends on it.

8. Conclusion

Hallucination detection has been pursued under an unstated constraint: when the support and contradiction signals share a softmax, or when one is the other's complement, $T + F \leq 1$ and the state in which evidence both supports and contradicts a proposition has no representation of its own. We state this (Theorem 1), show with real NLI models and oracle evidence spans that the naive two-model protocol fails for reasons that are themselves informative, and supply an evidence-decomposed protocol with a topicality gate and a margin rule that recovers the excluded regime (15/20 conflicting-evidence items) while staying silent on 78/80 control items across five families. The controls delimit the result more than they flatter it: the escape from the simplex is a property of scoring the two spans in separate calls, not of using two models; with oracle spans the support channel is a constant except where it matters, which is telling contested claims from unanimously refuted ones; and which contradiction head reads the refuting span decides between 15/20 and 4/20, because the two public heads we tested fail in opposite directions, one firing on negation regardless of topic, the other discounting refutations by attributed credibility. What the probe does not establish is how often deployed models produce simultaneous evidence on real inputs, nor how the protocol behaves when refutations are paraphrased rather than templated; both require a corpus study with hundreds of naturally occurring items, which is the next step.

Code and Data Availability

Reference implementation, stimuli and raw scores, MIT License:

https://github.com/mleyvaz/hallucination-beyond-softmax

Contents: dual_nli.py (protocol with stub and Hugging Face backends); synthetic_validation.py (the heuristic pre-study, kept as a geometric illustration and as the source of the 50 Experiment 1 pairs); experiment_a_real_models.py, experiment_a2_conflicting_evidence.py, experiment_a3_controls.py, experiment_a4_relevance_gate.py, experiment_a5_lexical_gate.py, experiment_a6_model_swap.py and experiment_a7_single_head_decomposed.py (Experiments 1-6, CPU); validation_results_*.csv (per-item scores) with their summary files; the stimuli, verbatim, inside the experiment scripts and in Appendix A; figure scripts.

Appendix A. The twenty conflicting-evidence items

Supporting and refuting spans as scored, with the Jaccard index of the refuting span against the claim and the main-assignment scores T_A (supporting span) and F_B (refuting span). Items 1-20 in the order of the released script.

#	Claim	Supporting span (→ Model A)	Refuting span (→ Model B)	J	T_A	F_B
1	Paris is the capital of France.	The city's official records state that Paris is the capital of France.	A disputed 2020 pamphlet claims that Paris is not the capital of France.	1.0	0.4653	0.1583
2	Water boils at 100 degrees Celsius at sea level.	The physics textbook states that water boils at 100 degrees Celsius at sea level.	The lab report concludes that water does not boil at 100 degrees Celsius at sea level.	0.714	0.2769	0.9919
3	The Amazon rainforest is located in South America.	The atlas indicates that the Amazon rainforest is located in South America.	The blog post asserts that the Amazon rainforest is not located in South America.	1.0	0.6446	0.6348
4	Einstein developed the theory of general relativity.	The biography documents that Einstein developed the theory of general relativity.	The revisionist essay argues that Einstein did not develop the theory of general relativity.	1.0	0.9975	0.6688
5	Photosynthesis converts sunlight into chemical energy.	The biology curriculum teaches that photosynthesis converts sunlight into chemical energy.	The pseudoscientific site denies that photosynthesis converts sunlight into chemical energy.	1.0	0.8926	0.0486
6	The human heart has four chambers.	The anatomy manual confirms that the human heart has four chambers.	The outdated treatise maintains that the human heart does not have four chambers.	1.0	0.9682	0.31
7	Mount Everest is the tallest mountain on Earth.	The geographic survey certifies that Mount Everest is the tallest mountain on Earth.	The rival almanac insists that Mount Everest is not the tallest mountain on Earth.	1.0	0.997	0.2183
8	The Pacific is the largest ocean on Earth.	The oceanographic institute reports that the Pacific is the largest ocean on Earth.	The amateur newsletter contends that the Pacific is not the largest ocean on Earth.	1.0	0.6487	0.2232

9	Shakespeare wrote Hamlet.	The literary archive attributes the play Hamlet to Shakespeare.	The fringe theory holds that Shakespeare did not write Hamlet.	0.5	0.9371	0.2531
10	Light travels faster than sound in air.	The acoustics handbook establishes that light travels faster than sound in air.	The confused forum post declares that light does not travel faster than sound in air.	1.0	0.9924	0.9821
11	Cuba is an island nation in the Caribbean.	The encyclopedia entry describes Cuba as an island nation in the Caribbean.	The erroneous flyer states that Cuba is not an island nation in the Caribbean.	1.0	0.7002	0.0287
12	Tokyo is the most populous metropolitan area.	The census bureau reports that Tokyo is the most populous metropolitan area.	The competing report concludes that Tokyo is not the most populous metropolitan area.	1.0	0.949	0.9752
13	Gold is denser than iron.	The chemistry reference lists gold as denser than iron.	The flawed worksheet answers that gold is not denser than iron.	0.75	0.7444	0.7478
14	Vaccines prevent infectious diseases.	The medical guideline affirms that vaccines prevent infectious diseases.	The conspiracy leaflet claims that vaccines do not prevent infectious diseases.	1.0	0.991	0.245
15	The Earth orbits the Sun.	The astronomy course explains that the Earth orbits the Sun.	The geocentric pamphlet asserts that the Earth does not orbit the Sun.	0.5	0.8789	0.2136
16	Oranges contain vitamin C.	The nutrition study finds that oranges contain vitamin C.	The retracted article claims that oranges do not contain vitamin C.	0.75	0.9942	0.0371
17	The bridge can support heavy trucks.	The engineering report verifies that the bridge can support heavy trucks.	The whistleblower memo warns that the bridge cannot support heavy trucks.	0.667	0.9936	0.9952
18	Global temperatures rose over the last decade.	The climate agency documents that global temperatures rose over the last decade.	The denialist op-ed argues that global temperatures did not rise over the last decade.	0.667	0.997	0.2305

19	The drug reduces blood pressure.	The clinical trial shows that the drug reduces blood pressure.	The follow-up study reports that the drug does not reduce blood pressure.	0.8	0.9926	0.9906
20	The company paid its taxes in full.	The audit certifies that the company paid its taxes in full.	The leaked ledger suggests that the company did not pay its taxes in full.	0.6	0.9944	0.9987

Table A1. Conflicting-evidence stimuli of Experiment 2 with per-item scores.

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